

1. a) Define Software Engineering and its significant role over the system design.

→ Software engineering is the discipline of designing, developing and maintaining software systems effectively and reliably.

→ In today's world, software has been increasingly embedded in various systems, from television set to air planes.

→ To deal with the increasing complex nature of the software, software engineering plays an important role.

→ Some of the major roles of SE are listed below:

i) SE ~~reas~~ researches, designs and develops software system to meet with client requirements.

ii) SE translates vague requirements and drives into precise specification.

iii) SE develops model for the system ~~and~~ or application.

iv) SE understands the behavior and performance of the system.

v) SE provides methods to schedule work & operate system at various levels and obtains the necessary details of the software.

vi) SE promotes interpersonal and communication skills and management skills.

1.b) Define agile software development with its advantages and disadvantages.

→ Agile software development is a methodology that values flexibility, collaboration and customer satisfaction.

→ Some advantages of agile software development are as follows:-

i) Agile is a very realistic approach for software development.

ii) Agile promotes team work and cross training.

iii) Agile is easy to manage.

iv) Agile gives flexibility to developers.

v) Agile enables concurrent development.

vi) Agile is suitable for fixed or changing requirements.

→ Disadvantages of agile software development are as follows:-

i) Agile is not suitable for handling complex dependencies.

ii) Agile has more risk of sustainability, maintainability and extensibility.

iii) Agile heavily depends on the customer interaction.

iv) Agile is quite challenging for the transfer of technology to new team members.

v) In agile, documentation and design are not given much attention.

2.a) Define risk. Explain how risk management is carried out in a software project? Explain all the steps.

sgf → Risk is an expectation of loss, a potential problem that may or may not occur in the future.

→ Risk management is the systematic process of identifying, addressing and eliminating the problems before they can damage the project.

→ Risk management is carried out in a software project by following steps:

i) Risk Identification:

→ Risk identification is the process of finding and documenting the potential risks that could affect the software project.

ii) Risk Analysis:

→ Risk analysis is the process of evaluating the likelihood and impact of each identified risk on the project's objectives, schedule, budget, quality or safety.

iii) Risk Mitigation:

→ Risk mitigation is the process of developing and maintaining implementing strategies and actions to reduce the risks and their negative effects on the software project.

iv) Risk Monitoring and Control:

→ Risk monitoring and control is the process of tracking and reviewing the risks and their management throughout the software project life cycle.

2.b) Explain why software system which is used in a real world environment must change or become progressively less useful.

Q. Software system which is used in a real world environment must change or become progressively less useful because of the following problems:-

i) Changing Requirements;

→ Software system must evolve according to user needs and business requirements.

ii) Technical Advancements;

→ Software system must emerge new technologies and platforms for better performance, security and features.

iii) Security Vulnerabilities;

→ Software system must be updated to address security vulnerabilities.

iv) Regulatory changes;

→ Software system must be updated to remain compliant with legal requirements.

v) Integration with other systems;

→ Software system must be updated to ensure compatibility and maintain seamless integration.

vi) Bug Fixes and Maintenance;

→ Software system must be updated to fix bugs and maintain system stability.

4. a) What is integration testing? Differentiate between top-down and bottom-up integration testing.

Integration testing is a software testing technique that focuses on verifying the interactions and data exchange between different components or modules of a software application.

S.N.	Top-Down Integration Testing	Bottom-Up Integration Testing
i)	In top-down integration testing, system integration begins with top level modules.	In bottom-up integration testing, system integration begins with lowest level modules.
ii)	In this testing, the higher level modules are test first then the lower level modules are tested and then the modules are integrated accordingly.	In this testing, the lower level modules are tested first then the higher level modules are tested and then the modules are integrated accordingly.
iii)	In top-down integration testing, Stub works as a momentary replacement.	In bottom-up integration testing, Driver works as a momentary replacement.
iv)	Top-down integration testing is implemented on POP languages.	Bottom-up integration testing is implemented on OOP languages.
v)	Top-down integration testing is simple.	Bottom-up integration testing is complex.
vi)	Top-down integration testing works on big to small components.	Bottom-up integration testing works on small to big components.
vii)	Top-down integration testing is more expensive.	Bottom-up integration testing is less expensive.

4.b) List out the differences between quality control and quality assurance and explain the elements and goals of software quality assurance (SQA).

S.N.	Quality Control (QC)	Quality Assurance (QA)
i>	Quality control focuses on full fulfilling the quality requested.	Quality Assurance focuses on providing assurance that the quality requested will be achieved.
ii>	Quality Control is the technique of verifying quality.	Quality Assurance is the technique of managing quality.
iii>	Quality Control is product oriented.	Quality Assurance is process oriented.
iv>	Quality Control aims to identify defects in the system.	Quality Assurance aim to prevent defects in the system.
v>	Quality control is performed after the Quality Assurance.	Quality Assurance is performed before Quality Control.

- SQA stands for software Quality Assurance.
- SQA is defined as a planned and systematic approach to the evaluation of the quality of software product standards, processes and procedures.
- ~~SQA~~ The elements of software Quality Assurance (SQA) are as follows:-
 - i> standards
 - ii> Reviews and Audits
 - iii> Testing
 - iv> Error Analysis
 - v> Change Management
 - vi> Education
 - vii> Vendor Management
 - viii> Security Management
 - ix> safety
 - x> Risk Management

→ The goals of SQA are as follows:

- i> Requirements Quality
- ii> Design Quality
- iii> Code Quality
- iv> Quality Control Effectiveness

5a) What is SPI (Software Process Improvement) and SPI framework? Explain briefly about CMMI.

sqf → Software Process Improvement (SPI) is a set of activities aimed at enhancing the quality and efficiency of software development processes.

→ SPI framework is a structured model designed to guide organizations in improving their software development processes.

→ SPI process is a structured model aimed at enhancing the effectiveness and efficiency of software development practices.

*CMMI:

→ CMMI stands for Capability Maturity Model Integration.
→ CMMI is a process model that guides organization to improve processes, boosts productivity and efficient behaviors during a project, product or service development which results in risk reduction.

→ According to the CMMI model, organizational process maturity is divided into 5 levels.

- i> Maturity Level 1: Initial
- ii> Maturity Level 2: Managed
- iii> Maturity Level 3: Defined
- iv> Maturity Level 4: Quantitatively managed
- v> Maturity Level 5: Optimizing

5.6> Define requirement engineering. What are the various methods of requirement elicitation? Explain.

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- Requirement engineering is the process of gathering the software requirements from the client, analyze and document them.
- Requirements elicitation is the process of investigating and learning about a system requirements from users, clients and other stakeholders.
- The various methods of requirement elicitation are as follows:
 - i> Interviews:
 - It is designed to understand the customer's expectations of the software.
 - ii> Brainstorming sessions:
 - It is designed to uncover new, innovative ideas and solutions.
 - iii> Prototyping:
 - It is designed to collect feedback from non-technical stakeholders by showing them an example.
 - iv> Observations:
 - It is designed to provide a direct view of, how a stakeholder performs a particular process.
 - v> Surveys:
 - It is designed to obtain feedback from large group of customers or end-users.
 - vi> Focus groups:
 - It is designed to get a lot of information at once.

7. Write short notes on:

a) Six Sigma:

- Six Sigma is a quality management methodology aiming to improve product and service quality by identifying and eliminating defects.
- The goal of Six Sigma is to deliver high-quality software that meets or exceeds customer expectations while reducing costs and improving efficiency.
- The five steps of Six Sigma methodology are as follows:
 - i) Define:
 - Six Sigma defines the goals of the project, the scope and the customer requirements.
 - ii) Measure:
 - Six Sigma collects data on current performance and defects.
 - iii) Analyze:
 - Six Sigma analyzes the data to identify root causes of defects.
 - iv) Improve:
 - Six Sigma develops and implements solutions to address the root causes.
 - v) Control:
 - Six Sigma implements controls to sustain the improvements.

b> Functional Independence:

- The functional independence is the concept of separation and related to the concept of modularity, abstraction and information hiding.
- The functional independence is accessed by using two criteria i.e. Cohesion and Coupling.

i> Cohesion:

- Cohesion is a measure of strength in relationship between various functions within a module.
- Cohesion is a natural extension of the information ~~hiding~~ ^{hiding} concept.

ii> Coupling:

- Coupling is a measure of strength in relationship between various modules within a software.
- Coupling is a measure of interconnection among modules in a program structure.

c> Deployment level design;

→ The deployment level design elements show the software functionality and subsystem that allocated in the physical computing environment which supports the software.

→ ~~Dep~~ Following figure shows three computing environment as shown. These are the personal computer, the CPI server and the control Panel.

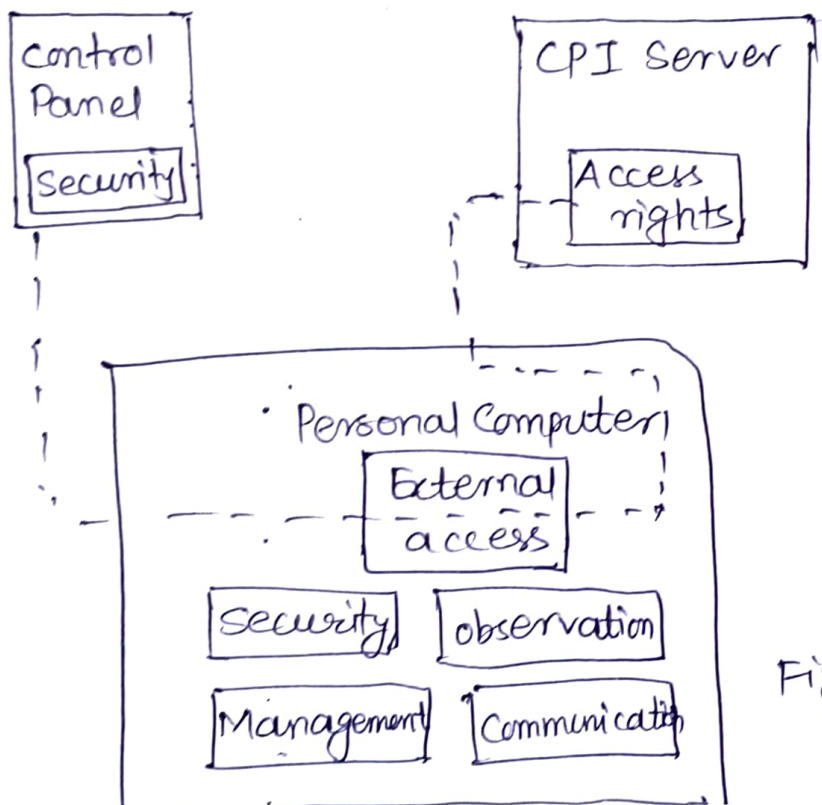


Fig. Deployment level diagram.